

COUNTEREXAMPLES TO THE GLUSHKOV MFN CONJECTURES

ABSTRACT. We refute the Glushkov parts of Conjectures 1 and 2 of Berglund, van der Merwe, and le Roux. For every $k \geq 2$, we give a regular expression for which the Minimum Feedback Node (MFN) set of the Glushkov automaton has size k , whereas the unique minimum memoization set is a singleton outside MFN. We characterize all memoization sets eliminating infinite degree of ambiguity and show that every ordering of IAR^+ returns the entire MFN set. Both output-to-optimum ratios therefore equal k and are unbounded. The corresponding Thompson questions are not addressed.

1. THE COUNTEREXAMPLE

Berglund, van der Merwe, and le Roux conjectured that, for every regular expression, a minimum memoization set can be chosen inside the Minimum Feedback Node (MFN) set, for both the Thompson and Glushkov constructions [1, Conjecture 1]. They equivalently conjectured that some ordering makes IAR^+ return a minimum memoization set [1, Conjecture 2 and Observation 3]. We disprove both Glushkov assertions.

We use the conventions of [1], in particular that $+$ is a primitive regular-expression operator. For an NFA A and $M \subseteq Q(A)$, write $A[M]$ for the corresponding memoized NFA. By the IDA characterization in [1, Theorem 1], $A[M]$ has infinite degree of ambiguity (IDA) if and only if there are distinct states p, q and a nonempty word v such that

$$p \xrightarrow{v} p, \quad p \xrightarrow{v} q, \quad q \xrightarrow{v} q,$$

with the q -cycle containing no state of M . Define

$$\mu(A) = \min\{|M| : M \subseteq Q(A), A[M] \text{ has no IDA}\}.$$

For an ordering $\prec$ of $\text{MFN}(A)$, let $\text{IAR}_{\prec}^+(A)$ denote the output of the procedure in [1, Definition 5].

For $k \geq 2$, let

$$R_k = \left(\underbrace{(a^+ \mid \cdots \mid a^+)}_{k \text{ occurrences}} b \right)^*, \quad A_k = \text{Gl}(R_k).$$

The alternatives are distinct syntax-tree occurrences and hence distinct positions in the Glushkov linearization; no algebraic simplification is applied

2020 *Mathematics Subject Classification.* 68Q45.

Key words and phrases. regular expressions, Glushkov automata, memoization, ambiguity, feedback vertex set.

before the construction. Write a_i^+ for the i th alternative. Let $x_1, \dots, x_k$ be the positions labelled a , let y be the position labelled b , and put

$$X_k = \{x_1, \dots, x_k\}.$$

Besides the initial state q_0 , the nonempty transitions are

$$\delta(q_0, a) = X_k, \quad \delta(x_i, a) = \{x_i\}, \quad \delta(x_i, b) = \{y\}, \quad \delta(y, a) = X_k$$

for $1 \leq i \leq k$. The final states are q_0 and y , and all states are useful.

Theorem 1. *For every $k \geq 2$ and every $M \subseteq Q(A_k)$,*

$$A_k[M] \text{ has no IDA} \iff y \in M \text{ or } X_k \subseteq M.$$

Moreover,

$$\text{MFN}(A_k) = X_k, \quad \mu(A_k) = 1,$$

and $\{y\}$ is the unique minimum memoization set. For every ordering $\prec$ of X_k ,

$$\text{IAR}_{\prec}^+(A_k) = X_k.$$

Proof. Algorithm 2 of [1] gives

$$a_i^+ \mapsto (\emptyset, \{x_i\}, \text{false}), \quad (a_1^+ \mid \dots \mid a_k^+) \mapsto (\emptyset, X_k, \text{false}).$$

Concatenation with b preserves the latter tuple, and the outer star changes only its nullability. Thus the root tuple is

$$(R_{\text{op}}, R_{\text{req}}, \text{nullable}(R_k)) = (\emptyset, X_k, \text{true}),$$

so

$$\text{MFN}(A_k) = X_k.$$

Independently, every feedback vertex set contains all of X_k , since each x_i has a self-loop, while deleting X_k leaves no directed cycle.

Suppose first that $y \in M$. Every directed cycle avoiding M is then a self-loop at some x_i , and every word labelling such a cycle is a^r for some $r \geq 1$. If a state p returns to itself on a^r , then $p = x_j$ for some j , and

$$\delta(x_j, a^r) = \{x_j\}.$$

Thus

$$p \xrightarrow{a^r} q \quad \text{and} \quad q \xrightarrow{a^r} p$$

force $p = q$, so the IDA criterion cannot hold. If instead $X_k \subseteq M$, every directed cycle meets M , and again there is no IDA.

Conversely, suppose that

$$y \notin M \quad \text{and} \quad X_k \not\subseteq M.$$

Choose $x_j \in X_k \setminus M$ and an index $\ell \neq j$. With

$$p = x_\ell, \quad q = x_j, \quad v = ba,$$

we have

$$x_\ell \xrightarrow{ba} x_\ell, \quad x_\ell \xrightarrow{ba} x_j, \quad x_j \xrightarrow{ba} x_j.$$

The q -cycle visits only x_j and y , neither of which lies in M . The IDA criterion imposes no avoidance condition on the other two paths, so x_ℓ may itself lie in M . Hence $A_k[M]$ has IDA. This proves the characterization.

The empty set therefore fails, while $\{y\}$ succeeds, so

$$\mu(A_k) = 1.$$

The characterization also shows that every successful singleton is $\{y\}$, proving uniqueness.

Finally, IAR^+ starts from X_k . Removing any x_i produces $X_k \setminus \{x_i\}$, which contains neither y nor all of X_k and therefore has IDA. Thus no state is removable, regardless of the ordering. $\square$

Corollary 2. *The Glushkov parts of Conjectures 1 and 2 in [1] are false. In fact, for every $k \geq 2$ and every ordering $\prec$,*

$$\frac{|\text{MFN}(A_k)|}{\mu(A_k)} = \frac{|\text{IAR}_\prec^+(A_k)|}{\mu(A_k)} = k.$$

Thus both gaps are unbounded.

Proof. The unique minimum memoization set is $\{y\}$, whereas

$$y \notin \text{MFN}(A_k) = X_k.$$

Consequently, no minimum memoization set is contained in MFN. The theorem also shows that every ordering of IAR^+ returns X_k . The ratios follow from $|X_k| = k$ and $\mu(A_k) = 1$. $\square$

1.1. A second family. The same two failures occur with distinct terminal letters and a minimum memoization set of size 2. For $k \geq 2$, let

$$S_k = \left((a \mid \underbrace{a^+ \mid \cdots \mid a^+}_{k \text{ occurrences}}) (b \mid c) \right)^*, \quad G_k = \text{Gl}(S_k),$$

again constructed from the displayed syntax without algebraic simplification, so that the repeated alternatives are distinct positions. Let z_0 be the position of the plain a , let $z_1, \dots, z_k$ be the positions of the k occurrences of a^+ , let y_b and y_c be the positions labelled b and c , and put $Z_k = \{z_0, z_1, \dots, z_k\}$. Besides the initial state q_0 , the nonempty transitions are

$$\delta(q_0, a) = \delta(y_b, a) = \delta(y_c, a) = Z_k, \quad \delta(z_i, b) = \{y_b\}, \quad \delta(z_i, c) = \{y_c\}$$

for $0 \leq i \leq k$, together with $\delta(z_i, a) = \{z_i\}$ for $1 \leq i \leq k$. The final states are q_0 , y_b and y_c .

Here Z_k is the unique minimum directed feedback vertex set of G_k : each z_i with $i \geq 1$ carries a self-loop and hence lies in every feedback vertex set; deleting those k states leaves the two cycles $z_0 \rightarrow y_b \rightarrow z_0$ and $z_0 \rightarrow y_c \rightarrow z_0$, of which z_0 is the only common state; and deleting Z_k leaves q_0, y_b, y_c with no transitions among them. Since the MFN algorithm returns a minimum feedback vertex set [1, Theorem 4],

$$\text{MFN}(G_k) = Z_k, \quad |\text{MFN}(G_k)| = k + 1.$$

Memoizing $\{y_b, y_c\}$ eliminates IDA: every directed cycle avoiding that set is a self-loop at some z_j with $j \geq 1$, so its label is a^r with $r \geq 1$, and any state returning to itself on a^r is such a z_j ; as $\delta(z_j, a^r) = \{z_j\}$ is a singleton, the two states of the IDA criterion coincide. Conversely, suppose M eliminates IDA and $|M| \leq 2$. If $y_b \notin M$, then, since $|Z_k| = k + 1 \geq 3$, we may choose $q \in Z_k \setminus M$ and $p \in Z_k \setminus \{q\}$, and

$$p \xrightarrow{ba} p, \quad p \xrightarrow{ba} q, \quad q \xrightarrow{ba} q,$$

where the q -cycle visits only q and y_b , neither in M ; so $y_b \in M$, and the word ca gives $y_c \in M$ in the same way. Hence

$$\mu(G_k) = 2,$$

and $\{y_b, y_c\}$ is the unique minimum memoization set. It is disjoint from $\text{MFN}(G_k)$. The same ba witness applies to every proper subset of Z_k , since $y_b \notin Z_k$, so no state of the initial MFN set is removable and $\text{IAR}_{\prec}^+(G_k) = Z_k$ for every ordering $\prec$. Thus S_k refutes the same two Glushkov assertions, with

$$\frac{|\text{MFN}(G_k)|}{\mu(G_k)} = \frac{|\text{IAR}_{\prec}^+(G_k)|}{\mu(G_k)} = \frac{k+1}{2},$$

which is again unbounded. For $k = 2$ this is the five-position expression $((a \mid a^+ \mid a^+)(b \mid c))^*$.

REFERENCES

- [1] M. Berglund, B. van der Merwe, and I. le Roux, *Selective memoization for efficient backtracking regular expression matching*, Electron. Proc. Theor. Comput. Sci. **446** (2026), 1–17. doi:10.4204/EPTCS.446.1.